

Machine Learning

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1 Linear model	2

1 Linear model

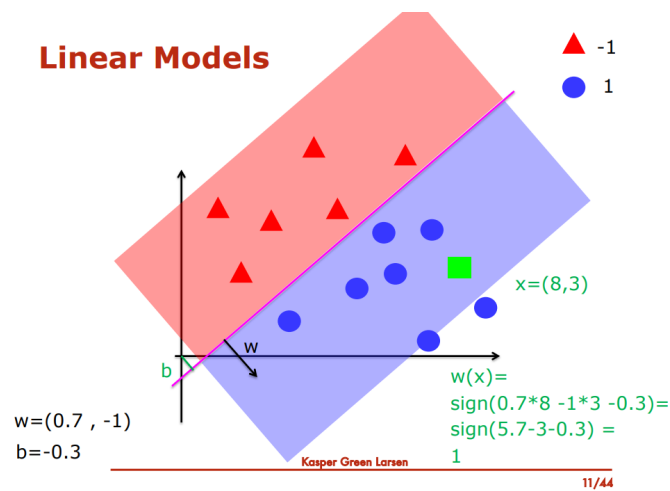
Hypothesis consists of the features vector:

$$x = \{x_1, \dots, x_d\}$$

and a bias b . If the linear combination of the weights and the input, and compare it to the bias, we can make predictions.

Example of mortgage, approve is 1 and decline is -1 . So we can predict that we approve

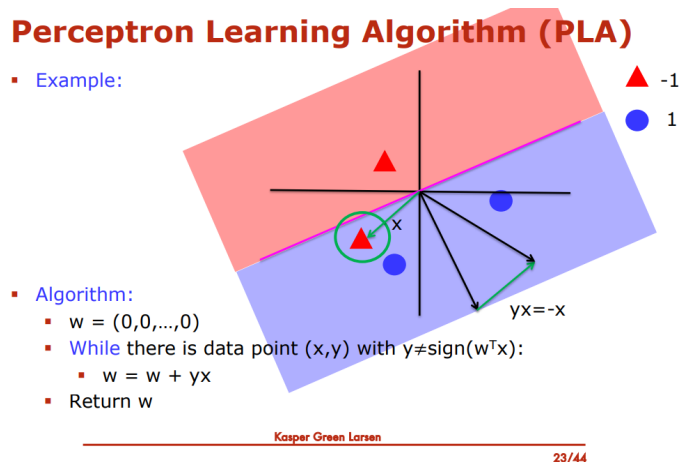
$$w_1x_1 + \dots + w_dx_d > -b$$



To find the parameters we can use **Perceptron Learning Algorithm (PLA)**.

Algorithm: While there is a data point (x, y) with

$$y \neq \text{sign}(w^T x) \rightarrow w = w + yx.$$



As we can see the algorithm can enter an infinite loop, as there might not be a linearly separable dataset.

A way to fix this is to use Pocket PLA which saves the most accurate hyperplane.

Pocket PLA

- Would like to still do something useful if not linearly separable!
- Pocket version: Remember best w in your pocket.
- Algorithm:
 - $w = (0,0,...,0)$
 - $bestScore = 0$
 - $bestW = w$
 - While you have more time, let (x,y) have $y \neq \text{sign}(w^T x)$:
 - $w = w + yx$
 - $score = \text{percentage of data correctly classified by } w.$
 - If $score > bestScore$:
 - $bestW = w, bestScore = score$
 - Return $bestW$